

Po-Chun Wu

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Education

Carnegie Mellon University

Master of Science in Electrical and Computer Engineering

Pittsburgh, PA

May 2028

Master of Science in Engineering and Technology Innovation Management (Dual Degree)

May 2028

Relevant Courses: Fundamentals of Computer Systems, Optimization, Networks in the Real World

National Taiwan University

Bachelor of Science in Electrical Engineering

Taipei, Taiwan

Jan 2026

GPA: 4.16/4.3

Relevant Courses: Data Structures, Algorithms, Machine Learning, SoC Verification, Computer Architecture

Projects

Formal Verification – NTU, Research Project (Team of 2)

Apr 2025 – Jun 2025

- Reduced the vocabulary dependency problem to an NP-hard Target Set Selection variant solvable via Boolean Satisfiability.
- Implemented a MiniSat-based bounded model checking algorithm to derive optimal solutions for graphs with 150 nodes.

LifeOS Automation System – NTU, Entrepreneurship Program (Team of 6)

Jan 2025 – Mar 2025

- Engineered a timestamp-based version control algorithm for a document-linking layer leveraging Asyncio in Python to resolve synchronization conflicts between multiple Notion databases and MongoDB records.
- Built a low-latency cache with a hybrid list-hash map for O(1) indexing, preventing redundant data fetches in MongoDB.

AR Navigation Helmet – Best Maker Award, 2024 MakeNTU Competition (Team of 5)

May 2024

- Created a navigation assistant mounted inside a motorcycle helmet, using speech recognition and AR to reduce distraction.
- Devised a multi-threaded Python system to handle asynchronous voice input, API requests, and OLED rendering.

NTU Pair – Personal Project

Dec 2022 – Feb 2023

- Developed a full-stack dating app with real-time messaging through *vanilla JS*, reaching 1,600+ verified users from NTU.
- Formulated a matchmaking algorithm to match users based on multi-dimensional psychological profiles.
- Rebuilt the system with secure coding practices after mitigating XSS, DoS, and credential-stuffing attacks.

Work Experience

SLEKMED

Taipei, Taiwan (Remote)

Product Developer (Part-Time)

Aug 2023 – Mar 2026

- Developed a web-based note-to-video generator integrating Marp, OpenAI API, ElevenLabs API, and FFmpeg, enabling medical experts to create high-quality educational videos from Markdown notes in less than 10 minutes.
- Designed a Next.js-based simulation module for medical diagnostic training in collaboration with Taipei Medical University peers, gaining 4,200+ followers and securing funding from the Taiwan Ministry of Education.
- Implemented an LLM-based chatbot as a virtual patient, contributing to a 1st-place finish among 40+ teams in the 2024 CTCL AI Competition (12K USD prize).

Skills

Programming: Python, JavaScript/TypeScript, C++, Verilog

Frameworks/Tools: React/Next.js, Node.js, Asyncio, MongoDB, MiniSat, Git

Specialties: System design, algorithm design, full-stack development, asynchronous programming, automated reasoning

Leadership

Founder and President, NTU Learning Optimization Club

Aug 2023 – Jun 2025

- Researched learning methodologies, delivered 8 nationwide speeches, and advised 200+ students on study habits.

Makerspace Administrator, NTUEE Student Association Academic Department

Sep 2023 – Dec 2023

- Organized 25 academic workshops and taught microcontroller programming to electrical engineering undergraduates.

Project Lead, NTUEE Virtual Makerspace

May 2023 – Aug 2023

- Led a 25-member team to build a React/Node.js inventory and reservation system for the university student makerspace.